

CONTROLLED TECHNICAL DOCUMENT

NAWI TEST REPORT

Type evaluation and verification record

DRAFT REPORT

TR-2026-000036

PASS

DEMONSTRATION / SYNTHETIC DATA

Synthetic observations are identified as such; reported results are the persisted application calculations.

REPORT AND INSTRUMENT IDENTIFICATION		
Report number TR-2026-000036	Document status DRAFT REPORT	Workflow status Awaiting review
Manufacturer NAWI Synthetic QA Lab	Applicant NAWI Synthetic QA Lab	Model / type designation Prototype Mechanical Bench III
Serial number QA3-III-001	Accuracy class / indication Class III / Non-self-indicating	Capacity 0.1 - 60 kg
e / d 0.02 / 0.01 kg	Tester Regression QA Tester Three	Reviewer samarth dubey
Applicable standard OIML R 76-1:2006	Revision 01	Issue date 25 Sept 2026, 03:00

APPLICANT AND MANUFACTURER

Field	Recorded information
Applicant contact	Regression QA Contact
Email	qa-applicant+qa3-001@local.test
Telephone	+91 9000000031
Applicant address	Synthetic QA Lab, Test Bench 3, Prototype Workspace.
Manufacturer address	Synthetic QA Lab, Test Bench 3, Prototype Workspace.

INSTRUMENT TECHNICAL SPECIFICATIONS

Field	Recorded information
Range / interval configuration	Single-Range / Single-Interval
Power source	Non-electrical instrument; no mains supply.
Tare configuration	No tare device
Stable equilibrium function	Yes
Multiple indicating devices	No
Printing capability	No
Data storage capability	No

TEST CONDITIONS

Parameter	Recorded start	Recorded end	Unit
Temperature	20	20	deg C
Relative humidity	45	50	%
Atmospheric pressure	1,013	1,013	hPa

Field	Recorded information
Evaluation started	23 Sept 2026, 04:32
Evaluation completed	25 Sept 2026, 02:58
Test setup notes	Synthetic prototype / regression data. Environmental values are fixed synthetic test conditions, not physical readings.
Preparation notes	Synthetic prototype / regression data. Environmental values are fixed synthetic test conditions, not physical readings.

Field	Recorded information
Environmental notes	Synthetic prototype end-of-session conditions; modeled values.

TEST EQUIPMENT AND CALIBRATION REFERENCES

No test equipment or calibration reference information is recorded.

APPLICABLE TEST SUMMARY

Test ID	Test description	Status	Result	Applicability / dependency reason
A.4.2	Checking of zero	Completed	PASS	
A.4.3	Setting to zero before loading	NOT APPLICABLE	NOT APPLICABLE	A.4.3 applies to instruments with digital indication; this instrument is configured without digital indication.
A.4.4	Determination of weighing performance	Completed	PASS	
A.4.5	Multiple indicating devices	NOT APPLICABLE	NOT APPLICABLE	The instrument does not have multiple indicating devices.
A.4.6	Tare	NOT APPLICABLE	NOT APPLICABLE	No tare device is configured.
A.4.7	Eccentricity	Completed	PASS	
A.4.8	Discrimination	Completed	PASS	
A.4.9	Sensitivity of non-self-indicating instrument	Completed	PASS	
A.4.10	Repeatability	Completed	PASS	
A.4.11	Variation of indication with time	Completed	PASS	
A.4.12	Stability of equilibrium	Completed	PASS	
A.4.13	Portable weighbridge tests	NOT APPLICABLE	NOT APPLICABLE	The instrument is not a portable road-vehicle weighbridge.
A.5	Influence Factors	Completed	PASS	
A.6	Endurance test	Completed	PASS	

DETAILED TEST RESULTS

A.4.2 - CHECKING OF ZERO

TEST RESULT: PASS

Applicability APPLICABLE	Technical reference OIML R 76-1:2006 Annex A A.4.2	Started 23 Sept 2026, 04:40
Completed 23 Sept 2026, 04:42		

A.4.3 - SETTING TO ZERO BEFORE LOADING

TEST RESULT: NOT APPLICABLE

Field	Recorded information
Status	NOT APPLICABLE
Reason	A.4.3 applies to instruments with digital indication; this instrument is configured without digital indication.

A.4.4 - DETERMINATION OF WEIGHING PERFORMANCE

TEST RESULT: PASS

Applicability APPLICABLE	Technical reference OIML R 76-1:2006 Annex A A.4.4	Started 23 Sept 2026, 04:42
Completed 23 Sept 2026, 04:44		

Weighing observations

No.	Applied load / indication / Delta L	P / Error Ec	MPE	Result
1	0.1 / 0.1 / 0 kg	0.11 / -0 kg	+/-0.01 kg	PASS
2	6 / 6 / 0 kg	6.01 / -0 kg	+/-0.01 kg	PASS
3	10 / 10 / 0 kg	10.01 / -0 kg	+/-0.01 kg	PASS
4	15 / 15 / 0 kg	15.01 / -0 kg	+/-0.02 kg	PASS
5	40 / 40 / 0 kg	40.01 / -0 kg	+/-0.02 kg	PASS
6	60 / 60 / 0 kg	60.01 / -0 kg	+/-0.03 kg	PASS
7	54 / 54 / 0 kg	54.01 / -0 kg	+/-0.03 kg	PASS
8	45 / 45 / 0 kg	45.01 / -0 kg	+/-0.03 kg	PASS
9	30 / 30 / 0 kg	30.01 / 0 kg	+/-0.02 kg	PASS

No.	Applied load / indication / Delta L	P / Error Ec	MPE	Result
10	0.12 / 0.12 / 0 kg	0.13 / 0 kg	+/-0.01 kg	PASS

Technical observations and calculations

Parameter	Recorded value	Parameter	Recorded value
Supported	Yes	Observation Unit	kg
Zero reference - Unit	kg	Zero reference - Zero Indication	0 kg
Zero reference - Additional load (Delta L0)	0 kg	Zero reference - Calculated E0	0.01
Zero reference - Recorded	23 Sept 2026, 04:43		

A.4.5 - MULTIPLE INDICATING DEVICES

TEST RESULT: NOT APPLICABLE

Field	Recorded information
Status	NOT APPLICABLE
Reason	The instrument does not have multiple indicating devices.

A.4.6 - TARE

TEST RESULT: NOT APPLICABLE

Field	Recorded information
Status	NOT APPLICABLE
Reason	No tare device is configured.

A.4.7 - ECCENTRICITY

TEST RESULT: PASS

Method Load receptor with not more than four points of support	Applicability APPLICABLE	Technical reference OIML R 76-1:2006 Annex A A.4.7; calculation A.4.4.3
Started 23 Sept 2026, 04:45	Completed 23 Sept 2026, 04:46	

Position observations

Series / phase	Position Id	Label	Loading Area	Geometry
----------------	-------------	-------	--------------	----------

Series / phase	Position Id	Label	Loading Area	Geometry
Quarter 1 - POSITION_1	Position 1	Quarter 1	Approximately one quarter of the load-receptor surface.	quarter
Quarter 2 - POSITION_2	Position 2	Quarter 2	Approximately one quarter of the load-receptor surface.	quarter
Quarter 3 - POSITION_3	Position 3	Quarter 3	Approximately one quarter of the load-receptor surface.	quarter
Quarter 4 - POSITION_4	Position 4	Quarter 4	Approximately one quarter of the load-receptor surface.	quarter

Observations

Series / phase	Load L	Indication I	Additional load (Delta L)	Corrected error	Maximum permissible error	Compliance Result
Quarter 1 - POSITION_1	20 g	20 g	0 g	0 g	0.02 g	PASS
Quarter 2 - POSITION_2	20 g	20 g	0 g	0 g	0.02 g	PASS
Quarter 3 - POSITION_3	20 g	20 g	0 g	0 g	0.02 g	PASS
Quarter 4 - POSITION_4	20 g	20 g	0 g	0 g	0.02 g	PASS

Technical observations and calculations

Parameter	Recorded value	Parameter	Recorded value
Execution Supported	Yes	Support Point Count	4
Position Count	4	Zero Condition Confirmed	Yes
Test Load - Value	20 kg	Test Load - Unit	kg
Sketch - Type	four-quarter-normal-platform		

A.4.8 - DISCRIMINATION

TEST RESULT: PASS

Method A.4.8.1	Applicability APPLICABLE	Technical reference OIML R 76-1:2006 Section 3.8.1 / Section 3.8.2.1; Annex A A.4.8.1
Started 23 Sept 2026, 04:58	Completed 23 Sept 2026, 04:59	

Discrimination observations

Load stage	Target	Starting indication	Lower indication / Delta	Upper indication / Delta	Applied / restored load	d checks	Result
Min	0.1 kg	Not recorded	Not recorded	Not recorded	Not recorded / Not recorded	Not recorded / Not recorded	PASS
1/2 Max	30 kg	Not recorded	Not recorded	Not recorded	Not recorded / Not recorded	Not recorded / Not recorded	PASS
Max	60 kg	Not recorded	Not recorded	Not recorded	Not recorded / Not recorded	Not recorded / Not recorded	PASS

A.4.9 - SENSITIVITY OF NON-SELF-INDICATING INSTRUMENT

TEST RESULT: PASS

Method A.4.9	Applicability APPLICABLE	Technical reference OIML R 76-1:2006 Annex A A.4.9
Started 23 Sept 2026, 04:59	Completed 23 Sept 2026, 05:00	

Procedure stages

Series / phase	Label	Order
Zero load	Zero load	1
Maximum load	Maximum load	2

Technical observations and calculations

Parameter	Recorded value	Parameter	Recorded value
Procedure Confirmation - Normal Oscillation Confirmed	Yes	Procedure Confirmation - Confirmed At	23 Sept 2026, 04:59

A.4.10 - REPEATABILITY

TEST RESULT: PASS

Method Same Load Repeat Weighings	Applicability APPLICABLE	Technical reference OIML R 76-1:2006 Section 3.6.1 / Annex A A.4.10
Started 23 Sept 2026, 05:00	Completed 23 Sept 2026, 05:01	

Observations

Series / phase	Repetition	Actual Load	Indication	Additional load (Delta L)	Corrected error	Maximum permissible error
Trial 1 / Approximately 80% of Max	1	48 kg	48 kg	0 kg	-0 kg	0.03 kg
Trial 2 / Approximately 80% of Max	2	48 kg	48 kg	0 kg	-0 kg	0.03 kg

Series / phase	Repetition	Actual Load	Indication	Additional load (Delta L)	Corrected error	Maximum permissible error
Trial 3 / Approximately 80% of Max	3	48 kg	48 kg	0 kg	-0 kg	0.03 kg

Technical observations and calculations

Parameter	Recorded value	Parameter	Recorded value
Control Stage	VERIFICATION	Zero Reference E0	0.01
Observation Unit	kg	Procedure Confirmation - Automatic Zero On Confirmed	Yes
Procedure Confirmation - Unloaded Instrument Rest Confirmed	Yes	Procedure Confirmation - Confirmed At	23 Sept 2026, 05:00

A.4.11 - VARIATION OF INDICATION WITH TIME

TEST RESULT: PASS

Method A.4.11.1 + A.4.11.2	Applicability APPLICABLE	Technical reference OIML R 76-1:2006 Section 3.9.4 / Annex A A.4.11
Started 23 Sept 2026, 05:05	Completed 25 Sept 2026, 02:54	

Variation and zero-return observations

Checkpoint	Elapsed	Load	Indication	Interpolation Delta L / zero deviation	P	Temperature	Result
Creep: T0	0 min	60 kg	60 kg	0 kg	60.01 kg	20 deg C	PASS
Creep: T5	5 min	60 kg	60 kg	0 kg	60.01 kg	20 deg C	PASS
Creep: T15	15 min	60 kg	60 kg	0 kg	60.01 kg	20 deg C	PASS
Creep: T30	30 min	60 kg	60.01 kg	0 kg	60.02 kg	20 deg C	PASS
Creep: T60	60 min	60 kg	60 kg	0 kg	60.01 kg	20 deg C	PASS
Creep: T120	120 min	60 kg	60 kg	0 kg	60.01 kg	20 deg C	PASS
Creep: T180	180 min	60 kg	60 kg	0 kg	60.01 kg	20 deg C	PASS
Creep: T240	240 min	60 kg	60 kg	0 kg	60.01 kg	20 deg C	PASS
Zero return: Post-load zero-return	After 30 min	-	Before 0; after 0 kg	0 kg	-	-	PASS

Variation calculation summary

Creep result PASS	Creep Delta P (T0 to T30) 0.01 kg	Creep limit 0.03 kg	Creep temperature variation 0 deg C
Zero return result PASS	Zero return zero-return deviation 0 kg	Zero return limit 0.01 kg	Zero return zero tracking off Yes
Zero return readings (before / after) 0 / 0 kg	Zero return stabilization before: confirmed; after: confirmed		

A.4.12 - STABILITY OF EQUILIBRIUM

TEST RESULT: PASS

Method Stability Of Equilibrium	Applicability APPLICABLE	Technical reference OIML R 76-1:2006 Section 4.4.2 / Annex A A.4.12
Started 25 Sept 2026, 03:00	Completed 25 Sept 2026, 03:00	

Stability phase summary

Phase	Description	Applicability	Result	Key observation / reason
A.4.12.1	Stability documentation/ configuration review	APPLICABLE	PASS	Review the manufacturer documentation for the stable-equilibrium principle, criteria, parameters, and security.
A.4.12.2	Printing / data-storage stability	NOT_APPLICABLE	Not determined	Not applicable - neither printing nor data storage is configured.
A.4.12.3	Zero-setting stability	APPLICABLE	PASS	Check inhibition before stable equilibrium and reuse A.4.2.3 evidence once stable.
A.4.12.4	Tare-balancing stability	NOT_APPLICABLE	Not determined	Not applicable - tare capability is not configured.
A.4.12.5	Continuous-disturbance inhibition	APPLICABLE	PASS	Record whether every applicable function is inhibited while continuous disturbance remains active.
A.4.12.6	Mobile/vehicle branch	NOT_APPLICABLE	Not determined	Not applicable - the instrument is configured as stationary.

Observations

Series / phase	Repetition	Additional load (Delta L)	Unit	Disturbance Confirmed	Operation Attempted Immediately	Blocked Before Stable
Trial 1 / A.4.12.3 - Zero-setting stability	1	0.01 kg	kg	Yes	Yes	Yes
Trial 2 / A.4.12.3 - Zero-setting stability	2	0.01 kg	kg	Yes	Yes	Yes
Trial 3 / A.4.12.3 - Zero-setting stability	3	0.01 kg	kg	Yes	Yes	Yes
Trial 4 / A.4.12.3 - Zero-setting stability	4	0.01 kg	kg	Yes	Yes	Yes

Series / phase	Repetition	Additional load (Delta L)	Unit	Disturbance Confirmed	Operation Attempted Immediately	Blocked Before Stable
Trial 5 / A.4.12.3 - Zero-setting stability	5	0.01 kg	kg	Yes	Yes	Yes

Technical observations and calculations

Parameter	Recorded value	Parameter	Recorded value
Setup - Target Load - Value	30 kg	Setup - Target Load - Unit	kg
Setup - Actual Load - Value	30 kg	Setup - Actual Load - Unit	kg
Setup - Input Unit	kg	Setup - Recorded	25 Sept 2026, 03:00
Zero Setting - Required Repetitions	5	Continuous Disturbance - Continuous Disturbance Confirmed	Yes
Continuous Disturbance - Recorded	25 Sept 2026, 03:00		

A.4.13 - PORTABLE WEIGHBRIDGE TESTS

TEST RESULT: NOT APPLICABLE

Field	Recorded information
Status	NOT APPLICABLE
Reason	The instrument is not a portable road-vehicle weighbridge.

A.5 - INFLUENCE FACTORS

TEST RESULT: PASS

Applicability APPLICABLE	Technical reference OIML R 76-1:2006 Annex A A.5	Started 25 Sept 2026, 02:56
Completed 25 Sept 2026, 02:57		

Influence factor results

Factor	Result	Key value / observation
A.5.1	PASS	Use the 50/1000 tilt procedure because no level indicator or automatic tilt sensor is configured.
A.5.2	NOT_APPLICABLE	A.5.2 is not applicable because the instrument does not use electric power.
A.5.3	PASS	The specified operating-temperature range is available for the A.5.3 procedure.
A.5.4	NOT_APPLICABLE	A.5.4 is not applicable because the instrument does not use electric power.

Observations

Indication	Additional load (Delta L)	Corrected error	Maximum permissible error	Load	Calculated indication
0	0.01	0	0.01	0	0
0.2	0.01	0	0.01	0.2	0.2
59.88	0.01	0	0.03	59.88	59.88
0	0.01	0	0.01	0	0
0.2	0.01	0	0.01	0.2	0.2
59.88	0.01	0	0.03	59.88	59.88

Technical observations and calculations

Parameter	Recorded value	Parameter	Recorded value
Setup - Environmental Conditions Stable Confirmed	Yes	Setup - Recorded	25 Sept 2026, 02:56
Tilting - Zero Setting Off Confirmed	Yes	Tilting - Recorded	25 Sept 2026, 02:56
Temperature - Recorded	25 Sept 2026, 02:57		

A.6 - ENDURANCE TEST

TEST RESULT: PASS

Applicability APPLICABLE	Technical reference OIML R 76-1:2006 Section 3.9.4.3 / Annex A A.6	Started 25 Sept 2026, 02:58
Completed 25 Sept 2026, 02:58		

Recovery checkpoints

Applications	Checkpoint time	Actual load	Recorded by	Notes
0	25 Sept 2026, 02:58	Not recorded	Regression QA Tester Seven	Session created; software checkpoint only.
10,000	25 Sept 2026, 02:58	Not recorded kg	Regression QA Tester Seven	Software recovery checkpoint; not an additional OIML interval.
25,000	25 Sept 2026, 02:58	Not recorded kg	Regression QA Tester Seven	Software recovery checkpoint; not an additional OIML interval.
50,000	25 Sept 2026, 02:58	Not recorded kg	Regression QA Tester Seven	Software recovery checkpoint; not an additional OIML interval.
75,000	25 Sept 2026, 02:58	Not recorded kg	Regression QA Tester Seven	Software recovery checkpoint; not an additional OIML interval.
100,000	25 Sept 2026, 02:58	Not recorded kg	Regression QA Tester Seven	Software recovery checkpoint; not an additional OIML interval.

Endurance calculation

Field	Recorded information
Required applications	100,000
Recorded applications	100,000
Pre-endurance baseline error	0 kg
Post-endurance corrected error	0 kg
Durability error	0 kg
Abnormal events	0
Cycle state	COMPLETED

Post-endurance indication

30 kg

REMARKS, DEVIATIONS AND NON-CONFORMITIES

Synthetic prototype / regression data. Environmental values are fixed synthetic test conditions, not physical readings.

Synthetic prototype end-of-session conditions; modeled values.

The first synthetic prototype zero-setting repetition used an I0 value that correctly produced 0.01 kg error against a 0.005 kg limit. This is an input-selection failure, not a calculation-engine defect. Retest with I0=0 and Delta L=0.5e, preserving the first calculated attempt.

OVERALL CONCLUSION

Overall evaluation result: PASS

The persisted evaluation record indicates that all applicable completed tests satisfy their recorded acceptance criteria.

REVIEW AND APPROVAL

Role	Name	Action / decision	Date
Tester	Regression QA Tester Three	Submitted for review	25 Sept 2026, 03:00
Reviewer	samarth dubey	Submitted for review	25 Sept 2026, 02:59

Reviewer note

The first synthetic prototype zero-setting repetition used an I0 value that correctly produced 0.01 kg error against a 0.005 kg limit. This is an input-selection failure, not a calculation-engine defect. Retest with I0=0 and Delta L=0.5e, preserving the first calculated attempt.

Review history

Action	Recorded by	Date
Submitted for reviewer assessment	Regression QA Tester Seven	25 Sept 2026, 02:58
Retest requested	samarth dubey	25 Sept 2026, 02:59

TRACEABILITY & REVISION

Technical basis OIML R 76-1:2006	Report format OIML R 76-2 structure	Revision 01
Document status DRAFT	Record basis Persisted tester observations and stored calculation results	Generated 25 Sept 2026, 03:00

EVIDENCE APPENDIX

1 persisted supporting record

Evidence 01 - A.4.8.1 - Discrimination - Test Setup

Field	Recorded information
Linked test	A.4.8 - A.4.8.1
Description	A.4.8.1 - Discrimination
Captured	23 Sept 2026, 05:15
Captured by	Regression QA Tester Three
Evidence type	discrimination_setup
Reference	A.4.8.1

